

Lecture 07: The VC Dimension

Measuring Model Capacity and Sample Complexity

Caltech CS156: Learning from Data

September 18, 2026

1 Definition of VC Dimension

In Lecture 06, we established that generalizability depends on the existence of a break point k . The **VC Dimension** formalizes this capacity into a single scalar metric.

Definition: VC Dimension d_{VC}

The VC Dimension of a hypothesis set $\mathcal{H}$, denoted $d_{VC}(\mathcal{H})$, is the maximum number of data points that can be **shattered** by $\mathcal{H}$.

$$d_{VC}(\mathcal{H}) = \max \{N \mid m_{\mathcal{H}}(N) = 2^N\} \quad (1)$$

If $d_{VC}(\mathcal{H})$ is finite, the smallest break point is $k = d_{VC}(\mathcal{H}) + 1$.

Substituting $k = d_{VC} + 1$ into Sauer's Lemma yields:

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^{d_{VC}} \binom{N}{i} \leq N^{d_{VC}} + 1 \quad (2)$$

2 VC Dimension of d -Dimensional Perceptrons

Theorem: d_{VC} for Linear Perceptrons

For a d -dimensional linear perceptron $\mathcal{H}$ (with $d + 1$ parameters including bias), $d_{VC}(\mathcal{H}) = d + 1$.

To prove $d_{VC} = d + 1$, we must prove two inequalities: $d_{VC} \geq d + 1$ and $d_{VC} \leq d + 1$.

2.1 Part 1: $d_{VC} \geq d + 1$ (Shattering $d + 1$ points)

Construct a matrix $\mathbf{X} \in \mathbb{R}^{(d+1) \times (d+1)}$ where the i -th row represents point $\mathbf{x}_i$ with $x_0 = 1$:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ 1 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & 1 \end{bmatrix} \quad (3)$$

Matrix $\mathbf{X}$ is lower-triangular with non-zero diagonal entries, hence $\mathbf{X}$ is invertible. For any arbitrary dichotomy vector $\mathbf{y} \in \{-1, +1\}^{d+1}$, we can solve $\mathbf{X}\mathbf{w} = \mathbf{y}$ directly as $\mathbf{w} = \mathbf{X}^{-1}\mathbf{y}$. Taking $\text{sgn}(\mathbf{X}\mathbf{w}) = \text{sgn}(\mathbf{y}) = \mathbf{y}$ realizes the exact dichotomy. Thus, $\mathcal{H}$ shatters this set of $d + 1$ points, proving $d_{VC} \geq d + 1$.

2.2 Part 2: $d_{VC} \leq d + 1$ (Cannot shatter $d + 2$ points)

Consider any set of $d + 2$ points in $\mathbb{R}^{d+1}$. Since there are $d + 2$ vectors in a $(d + 1)$ -dimensional vector space, they must be linearly dependent. Thus, there exist non-zero scalar coefficients $a_1, \dots, a_{d+2}$ such that:

$$\sum_{i=1}^{d+2} a_i \mathbf{x}_i = \mathbf{0} \quad (4)$$

Isolate a point with non-zero coefficient, say $\mathbf{x}_j$, so $\mathbf{x}_j = \sum_{i \neq j} b_i \mathbf{x}_i$ where $b_i = -a_i/a_j$. Now construct a dichotomy $\mathbf{y}$ where $y_i = \text{sgn}(b_i)$ for all $i \neq j$, and set $y_j = -1$. If a weight vector $\mathbf{w}$ realized this dichotomy, then $\mathbf{w}^T \mathbf{x}_i$ would share the sign of b_i , making $b_i(\mathbf{w}^T \mathbf{x}_i) > 0$. Taking the inner product with $\mathbf{w}$:

$$\mathbf{w}^T \mathbf{x}_j = \sum_{i \neq j} b_i(\mathbf{w}^T \mathbf{x}_i) > 0 \implies y_j = \text{sgn}(\mathbf{w}^T \mathbf{x}_j) = +1 \quad (5)$$

This contradicts $y_j = -1$. Thus, no linear perceptron can realize this dichotomy on $d + 2$ points, proving $d_{VC} \leq d + 1$. Combining both parts confirms $d_{VC} = d + 1$.

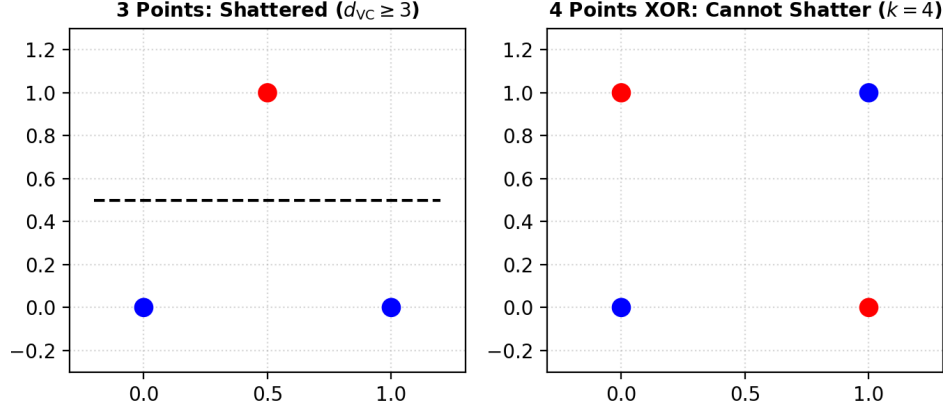


Figure 1: Linear classifier shatters 3 points in 2D ($d_{VC} \geq 3$), but cannot shatter 4 points (XOR pattern).

3 Interpreting d_{VC} : Degrees of Freedom & Sample Complexity

d_{VC} measures the **effective number of parameters** or physical degrees of freedom.

3.1 Generalization Error Bound

Setting the RHS of the VC inequality to δ , we derive the Generalization Bound with probability $\geq 1 - \delta$:

$$E_{\text{out}}(g) \leq E_{\text{in}}(g) + \underbrace{\sqrt{\frac{8}{N} \ln \left(\frac{4(2N)^{d_{VC}}}{\delta} \right)}}_{\Omega(N, \mathcal{H}, \delta) \text{ (Penalty for Complexity)}} \quad (6)$$

3.2 Practical Rule of Thumb for Sample Complexity

To achieve generalization error $\epsilon \approx 0.1$ with confidence 95% ($\delta = 0.05$), theoretical bounds require $N \approx 10,000 d_{VC}$. However, in practical engineering applications, the worst-case upper bound is overly conservative.

Practical Rule of Thumb

In practical machine learning workflows, a sample size of $N \geq 10 \times d_{VC}$ is typically sufficient to achieve good generalization performance.